COMPUTER PROGRAMMING

Electrical Engineering Department — UET Nowshera

CEP LAB

Lab Title: Relative Grading System Using UET Peshawar Formula

1. OBJECTIVE

The purpose of this CEP lab is to implement a Relative Grading System in C++ that calculates student GPA and letter grades using the official UET Peshawar grading formula. The program reads marks for multiple students, assigns initial grades, computes the class average, applies a relative grading offset to bring the class average to the UET target of 65, recalculates all grades after the shift, and identifies the class topper.

2. THEORY

UET Peshawar uses an absolute grading scale combined with a relative grading adjustment. In absolute grading, each student's letter grade and GPA points are determined directly from their marks out of 100 using fixed boundaries. In relative grading, the entire class's marks are shifted uniformly so that the class average reaches a target value (typically 65 at UET Peshawar). This prevents a situation where a difficult exam causes the whole class to fail unfairly.

UET Peshawar Grading Scale: - 85 and above — A — GPA 4.0
- 80 to 84 — A- — GPA 3.7
- 75 to 79 — B+ — GPA 3.3
- 71 to 74 — B — GPA 3.0
- 68 to 70 — B- — GPA 2.7
- 64 to 67 — C+ — GPA 2.3
- 61 to 63 — C — GPA 2.0
- 57 to 60 — C- — GPA 1.7
- 53 to 56 — D+ — GPA 1.3

- 50 to 52 — D — GPA 1.0
- Below 50 — F — GPA 0.0

Relative Grading Formula:

- Class Average = Sum of all marks / Number of students
- Offset = Target Average (65) - Class Average
- Adjusted Marks = Original Marks + Offset (capped between 0 and 100)

The program uses a struct called Student to store each student's name, marks, GPA, and letter grade. An array of this struct holds all students. Separate functions handle grade assignment, relative grading, average calculation (using a pointer), and finding the class topper (using a pointer). The reference parameter (&s) in assignGrade() allows the function to modify the struct directly.

3. SOFTWARE USED

- DEV-C++ IDE
- C++ Compiler (bundled with DEV-C++)

4. PROCEDURE

Task — Relative Grading System: UET Peshawar

Declare a structure called Student with members for name (char array), marks (float), GPA (float), and letter grade (char array). Write a function assignGrade() that uses a reference parameter to assign the correct letter grade and GPA to a student based on the UET grading scale using nested ifelse. Write a function calcAverage() that uses a pointer parameter to return the class average. Write a function applyRelative() that shifts all students' marks by the difference between the target average (65) and the actual class average, then reassigns grades. Write a function findTopper() that uses a pointer to return the index of the student with the highest marks. In main(), take student count and marks as input, display grades before and after relative grading, and print the class topper.

```
// ====================================================
// CEP: Relative Grading System — UET Peshawar
// Student: waqar khan  | Reg: BF25NWELE0720
// Semester 2 | UET Peshawar, Nowshera Campus
// ====================================================
```

```
#include <iostream>
#include <cstring>
using namespace std;


const int   MAX = 10;          // maximum students


struct Student                 // holds one student's data
{
    char  name[50];            // student name
    float marks;               // marks obtained (out of 100)
float gpa;                 // calculated GPA     char
grade[3];            // letter grade e.g. A, B+

};


void assignGrade(Student &s)   // assign grade by UET scale
{
    if      (s.marks >= 85) { s.gpa = 4.0; strcpy(s.grade, "A" ); }
else if (s.marks >= 80) { s.gpa = 3.7; strcpy(s.grade, "A-"); }     else
if (s.marks >= 75) { s.gpa = 3.3; strcpy(s.grade, "B+"); }

    else if (s.marks >= 71) { s.gpa = 3.0; strcpy(s.grade, "B" ); }
    else if (s.marks >= 68) { s.gpa = 2.7; strcpy(s.grade, "B-"); }
    else if (s.marks >= 64) { s.gpa = 2.3; strcpy(s.grade, "C+"); }
else if (s.marks >= 61) { s.gpa = 2.0; strcpy(s.grade, "C" ); }     else
if (s.marks >= 57) { s.gpa = 1.7; strcpy(s.grade, "C-"); }

    else if (s.marks >= 53) { s.gpa = 1.3; strcpy(s.grade, "D+"); }
else if (s.marks >= 50) { s.gpa = 1.0; strcpy(s.grade, "D" ); }
    else                    { s.gpa = 0.0; strcpy(s.grade, "F" ); }
}
```

```
void applyRelative(Student list[], int n, float classAvg, float
targetAvg)

{

    float offset = targetAvg - classAvg;  // marks shift amount
for (int i = 0; i < n; i++)

    {

        list[i].marks += offset;          // shift marks

        if (list[i].marks > 100) list[i].marks = 100;  // cap at 100

        if (list[i].marks < 0  ) list[i].marks = 0;    // floor at 0
assignGrade(list[i]);              // recalculate grade

    }

}
```

```
void calcAverage(Student list[], int n, float *avg)  // uses pointer

{

    float sum = 0;

    for (int i = 0; i < n; i++)

        sum += list[i].marks;

    *avg = sum / n;                        // store result via pointer
}


void findTopper(Student list[], int n, int *index)   // uses pointer
{

    *index = 0;

    for (int i = 1; i < n; i++)

        if (list[i].marks > list[*index].marks)
            *index = i;                    // update via pointer
```

```
}


int main()
{
    Student list[MAX];     int   n         = 0;
float targetAvg = 65.0;  // UET target average
```

```
    cout << "===== UET Peshawar Relative Grading System =====" << endl;
    cout  <<  "How  many  students?  (max  "  <<  MAX  <<  "):  ";
cin  >> n;


    if (n < 1 || n > MAX) {         cout << "Invalid. Enter
between 1 and " << MAX << endl;         return 0;     }




    for (int i = 0; i < n; i++)           // input loop
{
        cout << "\nStudent " << (i + 1) << ":" << endl;
        cout << "  Name  : ";        cin.ignore();

        cin.getline(list[i].name, 50);    // read name with spaces
cout << "  Marks : ";        cin >> list[i].marks;
assignGrade(list[i]);             // initial grade
    }
```

```
    float rawAvg = 0;     calcAverage(list, n, &rawAvg);
// pointer call

    cout << "\n--- Before Relative Grading ---" << endl;
    cout << "Class Average: " << rawAvg << endl;     for
(int i = 0; i < n; i++)
        cout << list[i].name << "  Marks: " << list[i].marks
                  << "  Grade: " << list[i].grade
                  << "  GPA: "   << list[i].gpa << endl;

    applyRelative(list, n, rawAvg, targetAvg);  // apply shift

    float newAvg = 0;
calcAverage(list, n, &newAvg);

    cout << "\n--- After Relative Grading ---" << endl;
cout << "Adjusted Average: " << newAvg << endl;
    for (int i = 0; i < n; i++)
        cout << list[i].name << "  Marks: " << list[i].marks
                  << "  Grade: " << list[i].grade
                  << "  GPA: "   << list[i].gpa << endl;

    int topIdx = 0;     findTopper(list, n, &topIdx);
// pointer call     cout << "\nClass Topper: " <<
list[topIdx].name
         << " (Marks: " << list[topIdx].marks
         << ", GPA: "   << list[topIdx].gpa << ")" << endl;

    return 0;
}
```

Output:

```
===== UET Peshawar Relative Grading System ===== How
many students? (max 10): 5


Student 1:
  Name  : Ahmad Khan


  Marks : 72
Student 2:
  Name  : Sara Bibi

  Marks : 58
Student 3:
  Name  : Usman Ali

  Marks : 45
Student 4:
  Name  : Fatima Noor


  Marks : 81
Student 5:
  Name  : Bilal Hassan

  Marks : 63



--- Before Relative Grading ---

Class Average: 63.8

Ahmad Khan   Marks: 72   Grade: B    GPA: 3
Sara Bibi    Marks: 58   Grade: C-   GPA: 1.7

Usman Ali    Marks: 45   Grade: F    GPA: 0


Fatima Noor  Marks: 81   Grade: A-   GPA: 3.7


Bilal Hassan Marks: 63   Grade: C    GPA: 2


--- After Relative Grading ---
```

```
Adjusted Average: 65

Ahmad Khan   Marks: 73.2  Grade: B    GPA: 3

Sara Bibi    Marks: 59.2  Grade: C-   GPA: 1.7


Usman Ali    Marks: 46.2  Grade: F    GPA: 0
Fatima Noor  Marks: 82.2  Grade: A-   GPA: 3.7


Bilal Hassan Marks: 64.2  Grade: C+   GPA: 2.3




Class Topper: Fatima Noor (Marks: 82.2, GPA: 3.7)
```

5. CONCLUSION

This CEP lab successfully implemented the UET Peshawar Relative Grading System in C++. A struct was used to store each student's name, marks, GPA, and letter grade together. An array of structures held all student records. The assignGrade() function used a reference parameter to directly update a student's grade and GPA using the official UET grading scale implemented through nested if-else statements. The calcAverage() and findTopper() functions used pointer parameters to return their results back to main(). The applyRelative() function shifted all marks uniformly to bring the class average to the UET target of 65, then recalculated grades for all students. The program correctly displayed results both before and after relative grading, and identified the class topper. All C++ concepts from Weeks 1 to 13 — variables, loops, if-else, functions, arrays, structs, pointers, and C-strings — were combined in this single practical program.